

1



Homology Search

jackhmmer search →
raw MSA

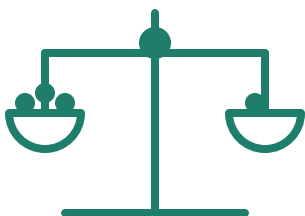
2

A C D ... Y
• • • ... -
• • • ... -
• • • ... -
• • • ... -

Clean & Filter MSA

Stockholm → N×L matrix,
EVEREST filters

3



Sequence Reweighting

99%-identity reweighting,
Neff

4



Build PSSM

per-column amino
acid frequencies

5

pos 1	0.5	-1.2		0.3
pos 2	-0.7	0.8		-1.1
...				
pos N	0.2	-0.4		1.3

Score Variants

log-odds scoring
of DMS variants

6



Evaluate

Spearman ρ vs. DMS assays,
bootstrap CI, scatter